

C++ Solutions — Pattern Practice Questions

Below are concise, ready-to-compile C++ solutions for each of the 10 pattern problems from your uploaded PDF. Each problem has its own `main()` so you can compile files individually or copy the code into separate `.cpp` files.

1) Right Triangle Star Pattern

```
#include <iostream>
using namespace std;
int main() {
    int N;
    if(!(cin>>N)) return 0;
    for(int i=1;i<=N;i++){
        for(int j=1;j<=i;j++){
            if(j>1) cout<<" ";
            cout<<"*";
        }
        if(i<N) cout<<"
";
    }
    return 0;
}
```

2) Inverted Right Triangle Star Pattern

```
#include <iostream>
using namespace std;
int main() {
    int N;
    if(!(cin>>N)) return 0;
    for(int i=N;i>=1;i--){
        for(int j=1;j<=i;j++){
            if(j>1) cout<<" ";
            cout<<"*";
        }
        if(i>1) cout<<"
";
    }
    return 0;
}
```

3) Half Pyramid of Numbers

```
#include <iostream>
using namespace std;
int main() {
    int N;
```

```

        if(!(cin>>N)) return 0;
        for(int i=1;i<=N;i++){
            for(int j=1;j<=i;j++){
                if(j>1) cout<<" ";
                cout<<j;
            }
            if(i<N) cout<<"
";
        }
        return 0;
    }
}

```

4) Inverted Number Pyramid

```

#include <iostream>
using namespace std;
int main() {
    int N;
    if(!(cin>>N)) return 0;
    for(int i=N;i>=1;i--){
        for(int j=1;j<=i;j++){
            if(j>1) cout<<" ";
            cout<<j;
        }
        if(i>1) cout<<"
";
    }
    return 0;
}

```

5) Floyd's Triangle Pattern

```

#include <iostream>
using namespace std;
int main() {
    int N;
    if(!(cin>>N)) return 0;
    int num = 1;
    for(int i=1;i<=N;i++){
        for(int j=1;j<=i;j++){
            if(j>1) cout<<" ";
            cout<<num++;
        }
        if(i<N) cout<<"
";
    }
    return 0;
}

```

6) Square Star Pattern

```

#include <iostream>
using namespace std;
int main() {
    int N;
    if(!(cin>>N)) return 0;
    for(int i=0;i<N;i++){
        for(int j=0;j<N;j++){
            if(j>0) cout<<" ";
            cout<<"*";
        }
        if(i<N-1) cout<<"
";
    }
    return 0;
}

```

7) Hollow Square Pattern

```

#include <iostream>
using namespace std;
int main() {
    int N;
    if(!(cin>>N)) return 0;
    if(N<=0) return 0;
    for(int i=1;i<=N;i++){
        for(int j=1;j<=N;j++){
            if(j>1) cout<<" ";
            if(i==1 || i==N || j==1 || j==N) cout<<"*";
            else cout<<" ";
        }
        if(i<N) cout<<"
";
    }
    return 0;
}

```

8) Pyramid Star Pattern

```

#include <iostream>
using namespace std;
int main() {
    int N;
    if(!(cin>>N)) return 0;
    for(int i=1;i<=N;i++){
        // leading spaces
        for(int s=1;s<=N-i;s++) cout<<" ";
        for(int j=1;j<=i;j++){
            if(j>1) cout<<" ";
            cout<<"*";
        }
    }
}

```

```

        if(i<N) cout<<"
";
    }
    return 0;
}

```

9) Diamond Star Pattern (N is half-height)

```

#include <iostream>
using namespace std;
int main() {
    int N;
    if(!(cin>>N)) return 0;
    // Upper half including middle
    for(int i=1;i<=N;i++){
        for(int s=1;s<=N-i;s++) cout<<" ";
        for(int j=1;j<=i;j++){
            if(j>1) cout<<" ";
            cout<<"*";
        }
        cout<<"
";
    }
    // Lower half
    for(int i=N-1;i>=1;i--){
        for(int s=1;s<=N-i;s++) cout<<" ";
        for(int j=1;j<=i;j++){
            if(j>1) cout<<" ";
            cout<<"*";
        }
        if(i>1) cout<<"
";
    }
    return 0;
}

```

10) Pascal's Triangle

```

#include <iostream>
#include <vector>
using namespace std;
int main() {
    int N;
    if(!(cin>>N)) return 0;
    for(int i=0;i<N;i++){
        long long c = 1;
        for(int j=0;j<=i;j++){
            if(j>0) cout<<" ";
            cout<<c;
            c = c * (i - j) / (j + 1);
        }
    }
}

```

```
        if(i<N-1) cout<<"  
";  
    }  
    return 0;  
}
```